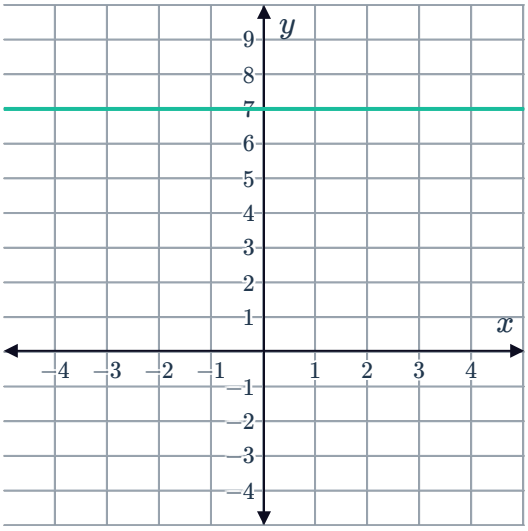
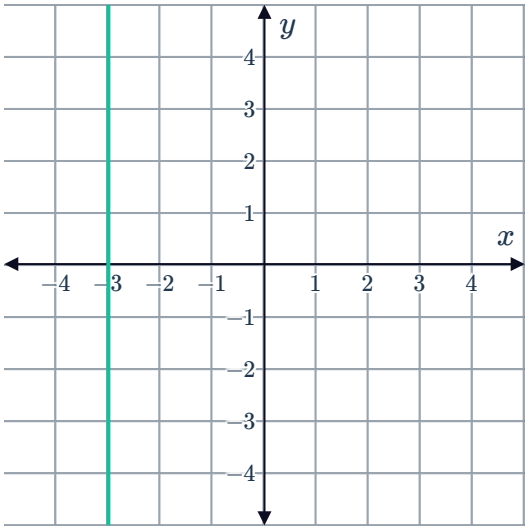
Gradient of a line

1 Describe the gradient of the lines in the following graphs:

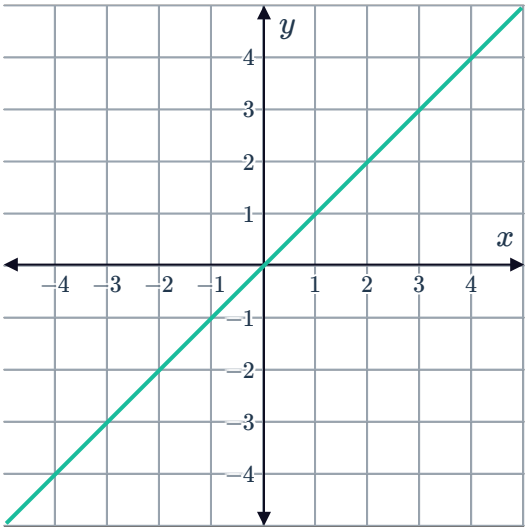
a



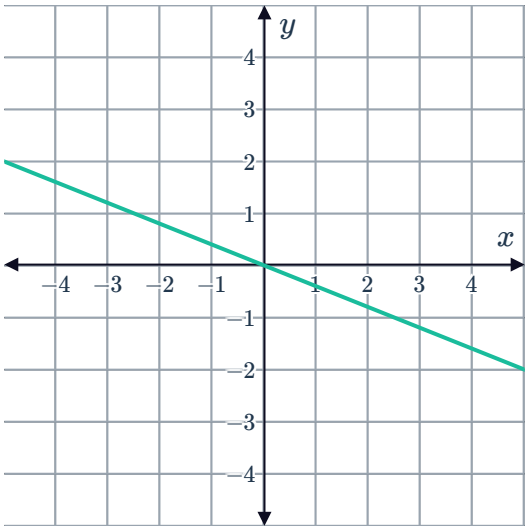
b



c



d



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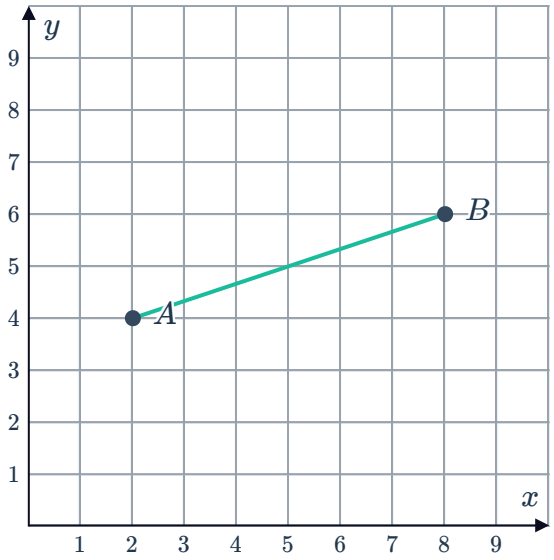
2 For each of the following intervals between points A and B:

i Find the rise.

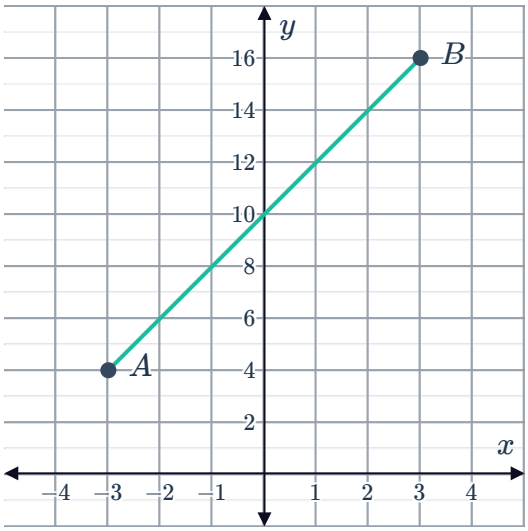
ii Find the run.

iii Find the gradient.

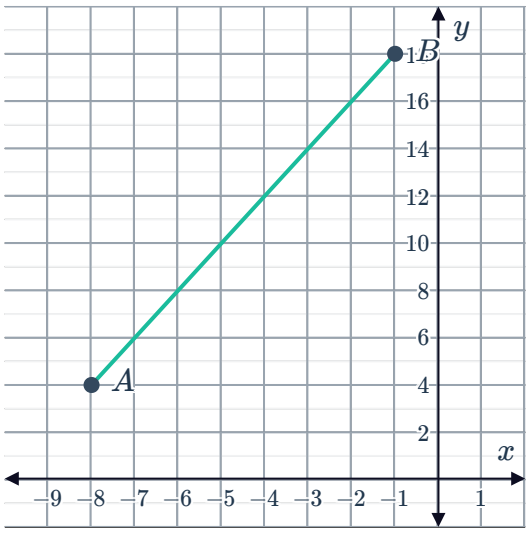
a $A\,(2,4)$ and $B\,(8,6)$



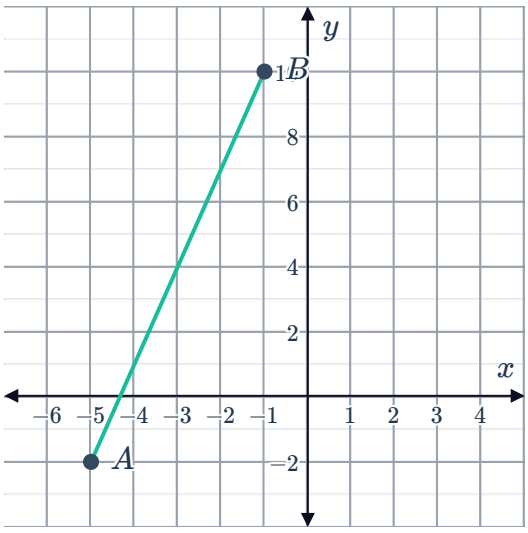
b $A\,(-3,4)$ and $B\,(3,16)$



c $A\,(-8,4)$ and $B\,(-1,18)$

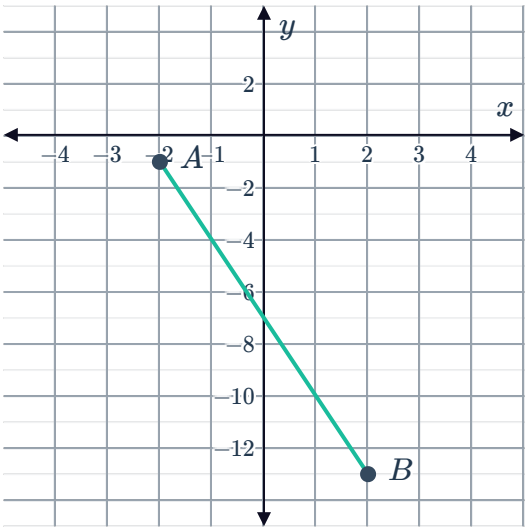


d $A\,(-5,-2)$ and $B\,(-1,10)$

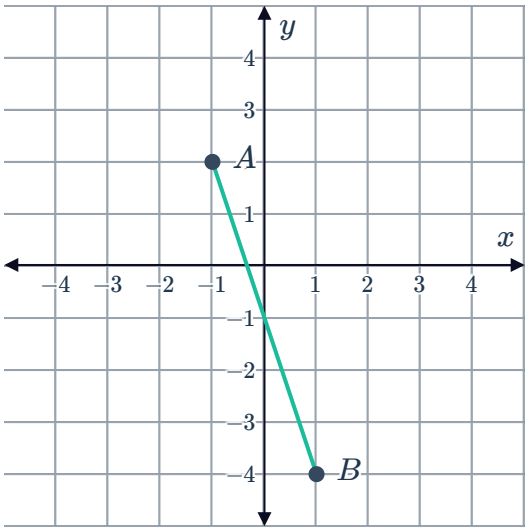


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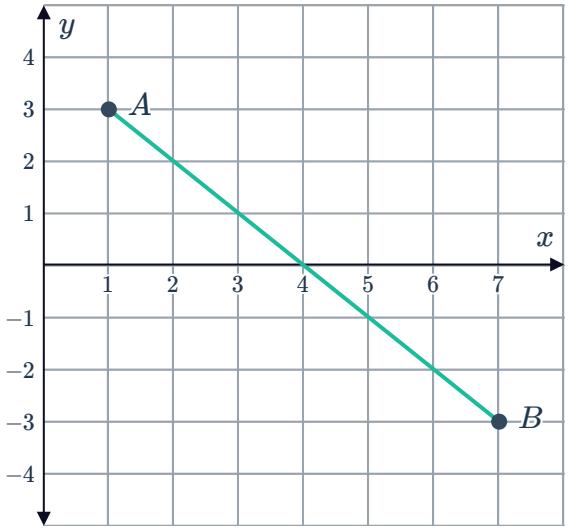
e $A\,(-2,-1)$ and $B\,(2,-13)$



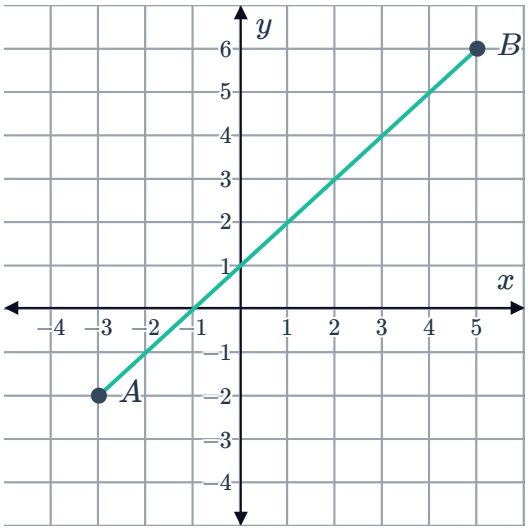
f $A\,(-1,2)$ and $B\,(1,-4)$



g $A\,(1,3)$ and $B\,(7,-3)$



h $A\,(-3,-2)$ and $B\,(5,6)$



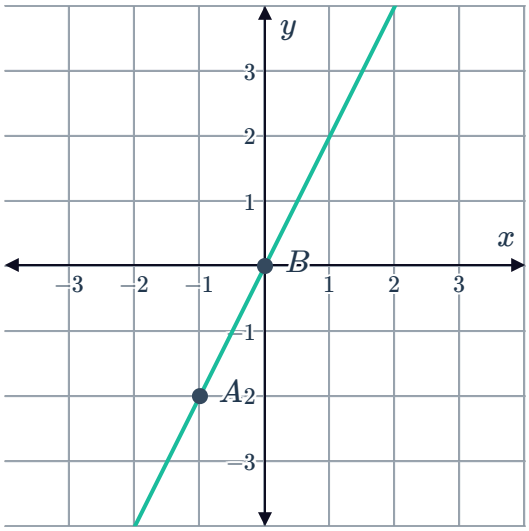
3 Explain why a vertical line has an undefined gradient.

4 State the gradient of any line parallel to the x-axis.

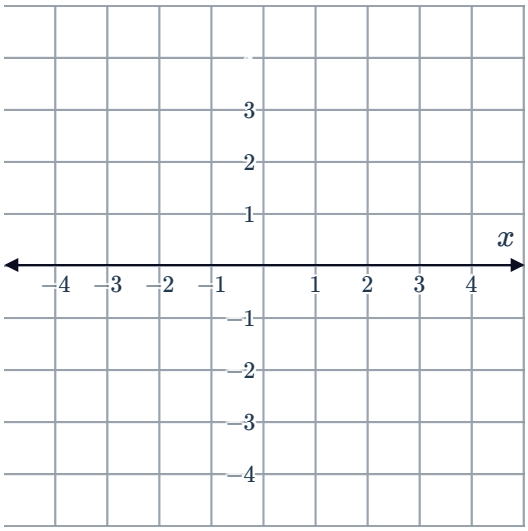
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5 Find the gradient of the following lines passing through points A and B:

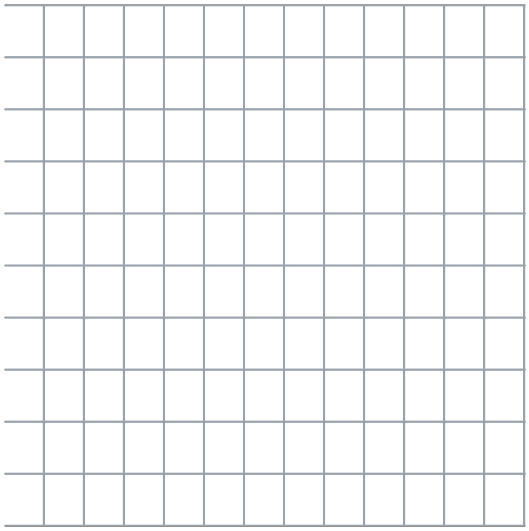
a



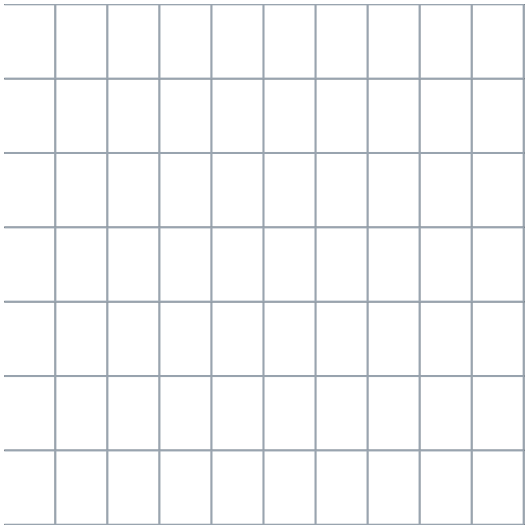
b



c



d



Gradient formula

6 Find the gradient of the interval joining $A\,(-9, 4)$ and $B\,(-3, -5)$.

7 If we have two points and the slope formula $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2)?